

Ansh Panjabi

Phone: +64 022 625 7807 | Email: ansh.panjabi99@gmail.com | [Linkedin](#) | [Portfolio/Website](#)

Profile Summary

Penultimate-year B.E. (Hons) Mechanical Engineering student combining hands-on automotive workshop experience with strong design and analysis capabilities. Proven track record applying FEA-driven design, CAD/CAM, and CNC machining to deliver precision-engineered components from concept through manufacture and physical testing.

Seeking Mechanical Engineering Internship (Nov 2026 – Feb 2027) to drive innovation and solve complex technical challenges.

Education

The University of Auckland

Bachelor of Engineering in Mechanical Engineering (Honours)

Auckland, New Zealand

Expected graduation, 2027

Related coursework: Finite element analysis, design and manufacture (CAD/CAM), control systems, materials mechanics, thermodynamics, fluid mechanics, dynamics, mechatronics, and engineering project management.

Leadership development: Completed the Faculty of Engineering Development Leadership Programme (Core); selected from hundreds of applicants to interview for DLP Elevate (2026).

Awards: University of Auckland Top Achiever Scholarship (2023); SVA Top Volunteer Award and Silver Service Award for 250+ hours of community service (2023).

Skills

Design & Analysis: FEA (ANSYS) for stiffness, strength and failure prediction, CAD modelling and dynamic simulation (Autodesk Inventor), CAM/NC toolpath programming (Inventor CAM, G-code), precision measurement tools (calipers, micrometers, torque wrenches)

Manufacturing & Workshop: CNC milling and turning, MIG welding, 3D printing, laser cutting, injection moulding, composite manufacturing, vehicle servicing, mechanical fabrication, hand/power tool operation

Professional: Technical reports and engineering drawings, design for manufacture, cross-functional teamwork, project leadership, event management, time management, safety protocols

Experience

Mechanical Intern | Bruce Mills 2012 Ltd

Jan 2026 – Feb 2026

- **Independently completed Warrant of Fitness inspections** and mechanical servicing across cars, vans, and trucks, including brake and brake cable replacements, oil and filter changes, and wheel servicing
- **Assisted senior mechanics on high-risk repairs** involving welding, structural work, and critical systems, gaining diagnostic exposure to cooling, electrical, and air/fuel delivery systems
- **Achieved zero safety incidents** while working largely unsupervised, applying engineering principles, workshop safety practices, and structured problem-solving to diagnose and resolve faults

Engineering Project Developer | University of Auckland

Mar 2024 – Present

- **Designed a lightweight aircraft-style structural bracket** using ANSYS FEA to meet 3.0 kN tension and 4.0 kN compression stiffness targets at minimum mass, verified by hand calculations and Instron testing
- **Programmed efficient CNC toolpaths** (Inventor CAM, G-code) for machining the bracket from polycarbonate sheet, designing for machinability within strict tooling and geometry constraints
- **Developed a multi-mechanism ball-transport device** to a 5.0 s timing target, with measured run times of 4.5 to 6.0 s across three runs; validated through Inventor dynamic simulation and prototyped with FDM 3D printing
- **Designed and injection-moulded a silicone phone case** with 3D-printed mould tooling, and fully documented a gear reducer design (shafts, gears, bearings, keys, clutch/brake sizing)
- **Software lead for a team robotics project**, developing the Arduino control code and integration for the ball pick-up and release, and contributing early CAD concepts

Leadership & Voluntary Activities

Structural Team Member | University of Auckland Rocketry Club

May 2026 – Present

Building high-powered rockets; composite manufacture of rocket structural components.

Events Executive | Startup Club UoA

April 2026 – Present

Lead end-to-end event planning and logistics connecting students with the startup ecosystem.

Events Executive | Auckland University Indian Society

Dec 2024 – Feb 2026

Coordinate cultural and social events, partnering with vendors and student leaders to drive engagement.

Youth Facilitator | Hindu Swayamsevak Sangh (NZ)

Jan 2021 – Apr 2023

Led week-long youth camps and community sessions promoting leadership and cultural education.